

Pick-to-Light System

Installation Guide

V.1.0

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1 Introduction

1.1 Product Overview

Pick-to-Light System is an advanced paperless picking system providing an innovative, streamlined and cost-effective solution to simplify the order fulfillment process in warehouses or distribution centers. Atop's Pick-to-Light System uses a state-of-the-art and light-directed technology to maximize the picking productively, speed and accuracy in different picking operation.

1.1.1 Features

- Ethernet Architecture
- Cableless design for easy installation and highest flexibility in location arrangement on shelf
- Versatility in different operating modes configuration
- Wide operating temperature range (-25°C~40°C)
- Certification: EMI, CE, FCC

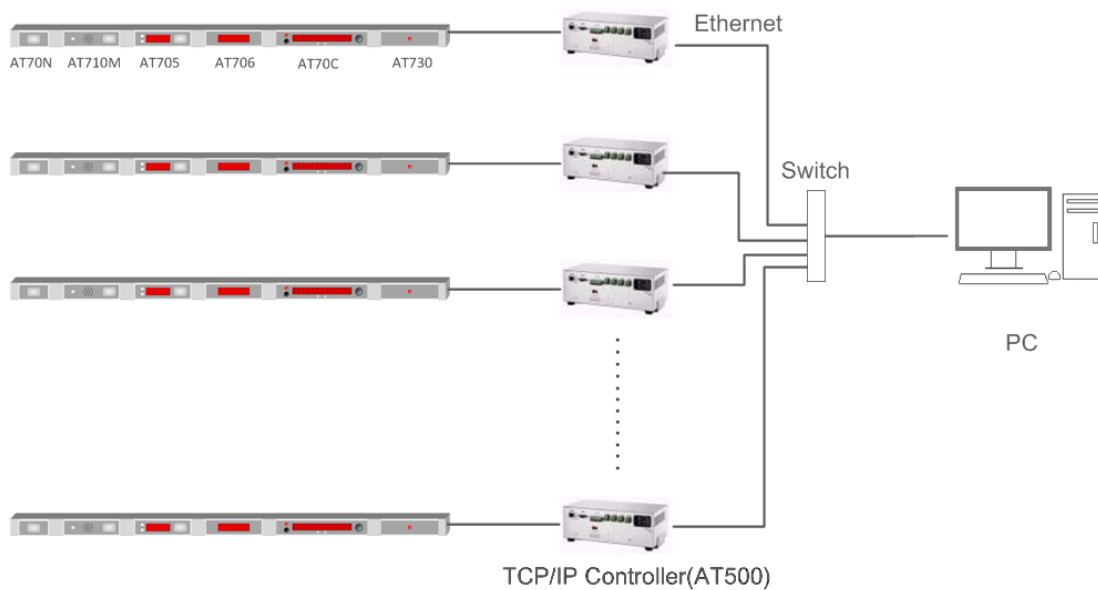


Figure 1.1

Using a switch to connect several controllers to the PTL system controlling PC

2 Wiring Schema

2.1 Planning a Wiring Schema for the Network

You will have to plan a wiring schema before installing the Pick-to-Light system in your warehouse/distribution center. You will need to figure out how many controllers (AT500) and picking tags will be needed and where they are going to be attached. You will also have to calculate how many rails and where they are going to be attached. After planning the wiring schema, you can start the installation by connecting to the networks. ABLEPick highly recommends that you create a closed network loop for the PTL system to prevent it from any interruptions with other networks. You can divide the network into two network loops by inserting two LAN cards in the PTL controlling PC. One is for the WMS/MIS server and the other one is for the PTL system server. Please refer to Figure 2.1 below.

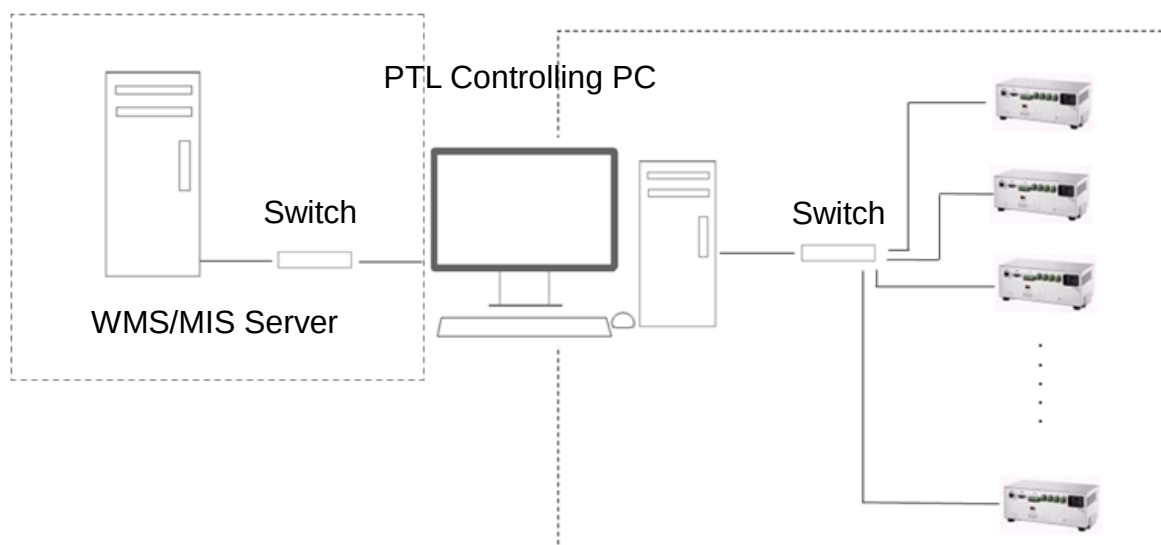


Figure 2.1

Dividing the network into two by inserting two LAN cards in the PTL controlling PC

3 Devices

3.1 TCP/IP Controller AT500



Figure 3.1

3.1.1 Features and Limitation

- One LAN port
- One AC input
- Full range of AC input is 110~230V
- Dimension: 250mm(L) x 130mm (W) x 100mm (H)
- Four output channels support up to 120 devices (tags)
- Maximum length of wiring in a bus should not exceed 30m

You can use switches to connect several controllers to the PC. Each controller (AT500) has four output channels, and each of it can support up to 30 devices, which means AT500 can support up to 120 devices (picking tags) at a time. as depicted in Figure 3.2.

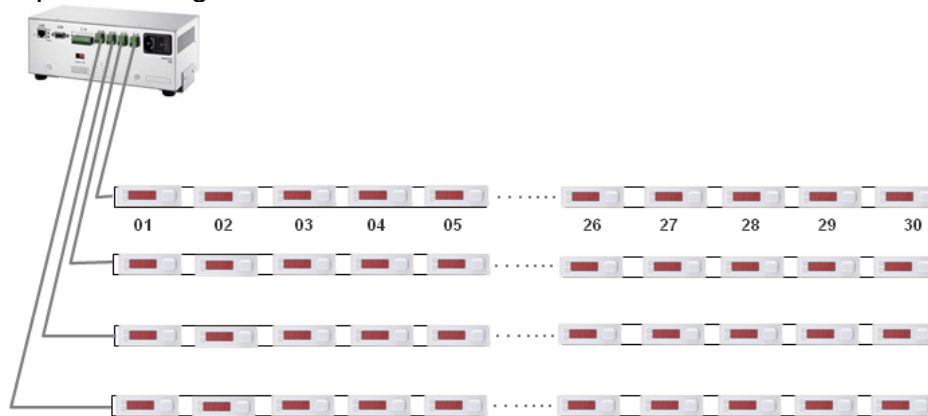


Figure 3.2

An AT500 can support up to 120 devices at a time.

3.2 Picking Tag AT705/L 5-Digit Pick Tag



Figure 3.3

3.2.1 5-Digit Display

- 6 LED colors (Red, Green, Orange, Blue, Pink, Cyan)
- 2 up/down function keys
- Power requirements: DC12V, Max: 100mA
- Size: 148mm (L) x 46mm (W) x 30mm (H)
- Operating temperature: 0°C~40°C (Frozen Model: -25°C~-40°C)

4 Installing the AT500

4.1 Find the Best Location to Mount the AT500

The best location to mount the AT500 on should be a place that can be easily connected to the internet and the AC power (should be grounded for safety.) Also, it should be easy to be maintained. You can place the AT500 into a cabinet to avoid humidity, and set an AC outlet (110V/220V) beside the AT500 within the cabinet to shorten the distance between AT500 and AC power. This is to reduce the redundancy of wiring and it will help you save space to put more devices (picking tags) on the main bus (maximum length of wiring in a bus should not exceed 30m.) Finally, please install the AT500 next to the devices as near as possible.



Figure 4.1

Mounting the cabinet on the rack



Figure 4.2

Setting an AC outlet beside the AT500

5 Materials

You will need the accessories and tools listed in this chapter:

5.1 Accessories

Rails, rail covers, end covers, a roll of conduction bar, terminal blocks, Velcro tape, edge protectors, screws, Y terminals, twin-wired cables, a cabinet and soft pipes.

REMARK: You can purchase all the items listed below from Atop, but you can purchase the star-marked() items locally by your own, too.*



※ Suggested spec

Screw: Head: 10.2 mm (Head Thickness: 2.5mm)/ Length: 10~12 mm

Velcro tape: 50 mm

5.2 Tools

You will need to prepare a circular saw with stand, electric drill, metal-cutting scissors, glue gun, screwdrivers, a pair of plier, wire stripper and a crimping tool.



Circular saw with stand



Electric drill



Metal-cutting scissors



Glue gun



Philips screwdriver



Flathead screwdriver



Plier



Wire stripper



Crimping tool



Vinyl adhesive*

6 Assembling the Rail

6.1 Rail

- Material: Plastic (PVC)
- Dimension: L: 2000 / 3000 x W:46 x H:25(mm)



6.1.1 Cutting the Rails into a Proper Length

ABLEPick provides two different kinds of specifications of rails, 2m and 3m. You can cut the rails into different lengths to fit your storage racks. But be aware that you have to reserve some lengths of rails for the end covers (4mm), and please always remember that the rails should be perfectly fit on the racks, so you need to shorten the rails to avoid attaching them to the ragged surfaces on the racks. (e.g. the welded points) The cut-edges of the rails should be cut in 90 angel degree so that the end covers can be perfectly fit on without any apertures.



Figure 6.1

You can raise the rail up the same level of the stand of the circular saw to cut in 90 angle degree.

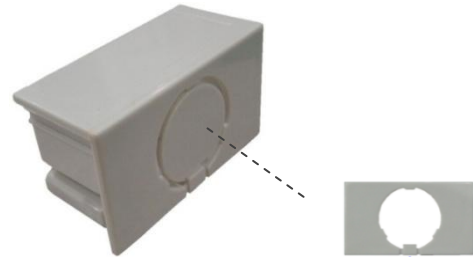


Figure 6.2

Cutting a bundle of rails fixed with gaffer tape

6.2 End Covers

- Material: Plastic (PVC)
- Dimension: L:20 x W:26 x H:27(mm)
- A disposable wiring hole



6.2.1 Put the End Covers on

After cutting the rails into sections, you may use common PVC glue to attach the end covers to the both ends of the rails. Please make sure the end covers are glued firmly. Vinyl adhesive would be a good choice for this process.

Follow the steps below to fix the end covers in the rail:

1. Glue two sides and the bottom of the end cover carefully.



2. Insert the end cover into the rail vertically, and wait for a second to let it dry.



6.3 Mounting the Rails on the Racks by Screws

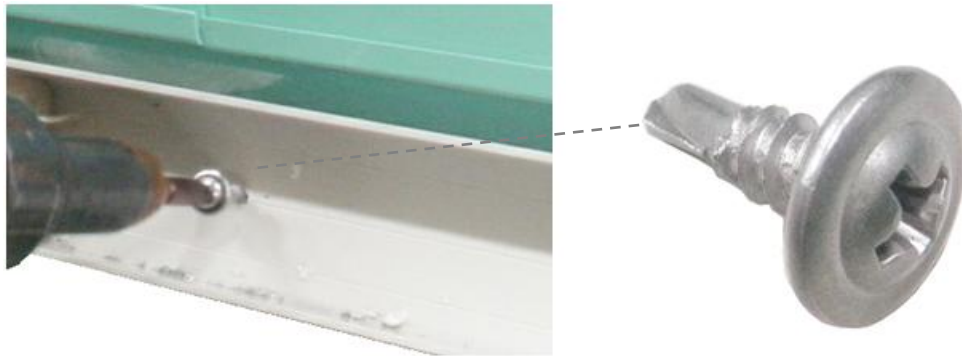


Figure 6.3

Using screws to mount the rails on the racks

Use some self-tapping screws to attach the rails to the racks by an electric drill or using bolts and nuts to prevent any injuries to the pickers in the distribution centers. There should be at least three screws driven in a rail. All screws in the rails should be covered by the conduction bars to prevent them from being touched by the cut-edges of the stainless steel strip on the conduction bar.



Figure 6.4

A wiring hole on the end cover



Figure 6.5

A wiring hole (Diameter 26mm) drilled in the rail

6.4 Conduction Bar

- Material: Plastic (PVC) and stainless steel strip
- L:100000 / 3000 x W:26 x H:2(mm)
- Packaged in 100m (a roll)



The conduction bar is the medium for power and communication from the controller (AT500) to the devices in the PTL system. You need to unroll the conduction bar first and make it as flat as possible to prevent any poor contacts to the devices. You will need a pair of metal-cutting scissors to cut the conduction bar into sections in proper lengths to fit in the rails.

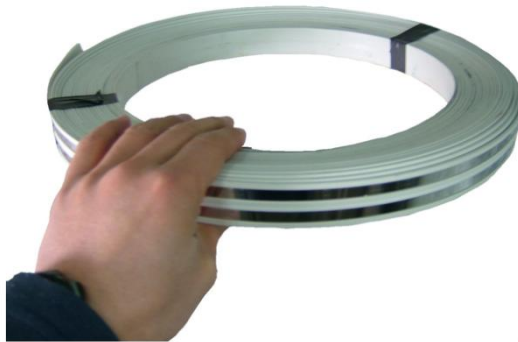


Figure 6.6

A roll of conduction bar (100m)



Figure 6.7

Cutting the conduction bar into sections with metal-cutting scissors

6.5 Edge Protectors

After cutting the conduction bars into sections, please seal all the cut-edges completely by using a glue gun or the edge protectors to avoid them to touch any metals.



Figure 6.8

Sealing the cut-edges of the conduction bar with the edge protectors



Figure 6.9

Sealing the cut-edges of the conduction bar with a glue gun

6.6 Velcro Hook and Loop Fastener

- L:50 x W:25 x H:4(mm)



You can either have Atop's Velcro tapes (5cm per set) or purchasing one locally by your own. You will need to put one loop side and the other hook side in the rail and on the back of the conduction bars respectively about every 2m. However, too many Velcro tapes will make the conduction bar sinking in the rail. This will hinder the connection between the conduction bar and the devices. You will have to wipe all the sawdust out of the rails with a dry cloth firstly, and stick a Velcro hook/loop in a rail by sticking a Velcro hook/loop within Line 1 to Line 3 in the rail, as indicated in Figure 6.10. Sticking a Velcro loop/hook on the back of the conduction bar (Figure 6.11) And, fix the conduction bar in the rail to complete this process (Figure 6.12).



Figure 6.10



Figure 6.11



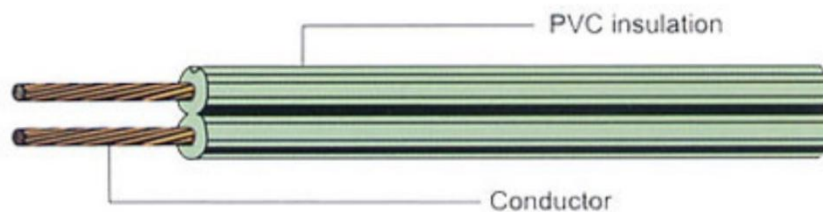
Figure 6.12

7 Wiring

After installing all the conduction bars in the rails on the storage racks, you may begin the wiring. You can use a twin-wired cable on which there are two different colors marked. This is to help you to distinguish the wires much easier. The twin-wired cable provides both electrical power and data connection from the AT500 through the terminal blocks to the conduction bar, and to the devices.

7.1 Twin-wired Cable

- Suggested specification: 1.25mm x 2C/ 0.18mm x 50C/ 16AWG



7.2 Terminal Block AT-790

- A/B connectors with screws
- Magnetic
- L:35 x W:26 x H:15(mm)

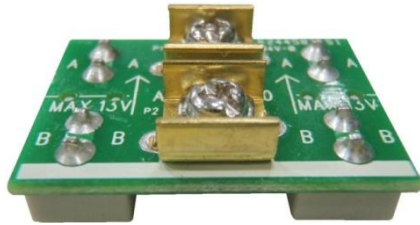


Figure 7.1



Figure 7.2

Always put the “A” connector toward the top, and the “B” connector toward the bottom on the conduction bar.

7.3 Locating the Terminal Blocks

You should stick a terminal block with the A connector toward to the top and the B connector toward to the bottom in the middle of a 2~3m rail. If you have a rail extending to 5~8m, you will need to put two terminal blocks on the rail. The best ratio to locate the terminal blocks on the rail is 1:2:1. Further, if you need up to three terminal blocks on a 5~8m rail, the ratio to locate the terminal blocks will be 1:2:2:1. Please refer to Figure 7.3.

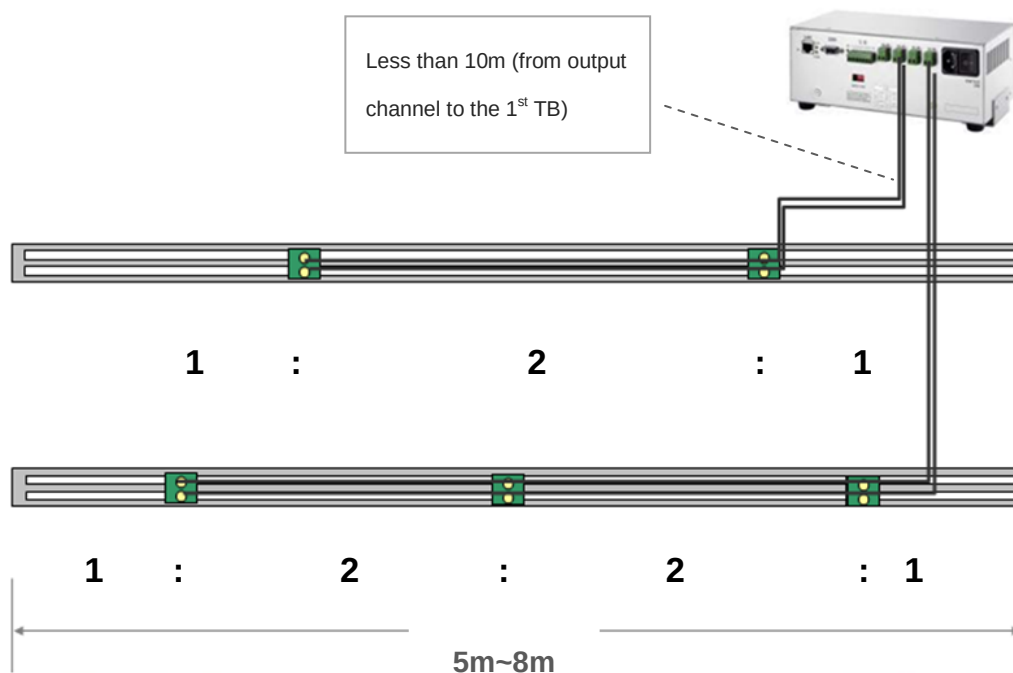


Figure 7.3

7.3.1 Making Y Connectors

You will need terminal blocks, twin-wired cable, Y terminals, wire stripper, and a crimper. You need to cut the twin-wired cable into a proper length and put the Y terminals on the end of the wires with a crimping tool to make Y connectors. Please refer Figure 7.4 as an example. First, you have to separate a twin-wired cable 30 mm off, and have its head 5mm stripped by using a wire stripper. The exposed red wire should be inserted into the red barrel of a Y terminal, and the exposed white wire will be inserted into a blue barrel of a Y terminal for better identifying later. They should pass through the barrels to touch the metal just a little. And then place each of the Y terminals with the wires in the notch 5.5mm on the crimping tool and crimp barrels to make Y connectors.



Figure 7.4

7.4 Wiring the AT500 to the Rail

There are four sets of output channels on the AT500: 1A1B, 2A2B, 3A3B and 4A4B. While wiring the AT500 to the rail, you need to make Y connectors (refer to Figure 7.4) for one of the end of the twin-wired cable, and connect the “red” Y connector to the “A” connector on the terminal block (should be toward the top on the conduction bar), and also have the “blue” Y connector connected to the “B” connector on the terminal block. Please connect them recursively to prevent them from being pulled out of the terminal block easily (for more information please refer to 7.4.2.) Besides, you need to have the other end of the wire 30mm separated, 5mm stripped and twisted at the head, and then inserted them through the holes of the green terminal respectively which correspond A and B output channels on the AT500. Take Figure 7.5 and Figure 7.6 as an example, please have the “red” twisted wire inserted through the hole of the green terminal which corresponds with “1A” output channel on the AT500, and the other “white” twisted one inserted through the hole of the green terminal which corresponds with “1B” output channel on the AT500, and tighten both screws on the top of the green terminal securely with a flathead screwdriver.



Figure 7.5



Figure 7.6

Insert the “red” twisted wire through the hole of the green terminal corresponding with “1A” output channel on the AT500, and insert the other “white” twisted one through the hole of the green terminal corresponding with “1B” output channel on the AT500, and tighten the screws securely on the top of the green terminal by using a flathead screwdriver.

7.4.1 Wiring with the Y Connectors

After finishing wiring the AT500, please wire the rest of the terminal blocks with the twin-wired cable. Please make Y connectors on the both ends of the twin-wired cable, and connect the “red” Y connector to the “A” connector recursively on the terminal block which should be placed toward the top on the conduction bar, and the “blue” one to the “B” connector on the terminal block which should be placed toward the bottom. The wiring for each of the terminal blocks in the whole PTL system has to be connected consistently or an unstable communication may occur.

7.4.2 Connecting Recursively

ABLEPick suggests you to wire the TBs recursively to prevent the wires from being pulled out of the terminal blocks easily, so please remember to reserve lengths for the sections of the wiring; for example, reserve 10 cm more for the first section of the wire (AT500 to 1st terminal block), and 20 cm more for the rest of the sections of wires (TBs to TBs). Please refer to Figure 7.7.

TCP/IP Controller (AT500)

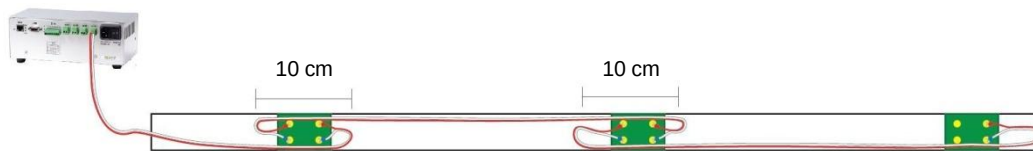


Figure 7.7

Attaching the terminal blocks to the conduction bar and connect the Y connectors recursively to prevent the wires from being pulled out of the terminal blocks easily.

7.4.3 Maximum Length of Wiring in a Bus Should not Exceed 30m

Be known that the total length of the whole wiring for each of the bus in the PTL system should not exceed 30m. The total length consists of the distance between the output channel on the AT500 to the first terminal block (TB), the first TB to the second TB, and the second TB and the third TB, and so on. It is very important to keep the whole wiring less than 30m in a bus to prevent the voltage from dropping with distance. Also, it will be better to keep the distance from the output channel to the first TB less than 10m to reduce the redundancy of wiring. Besides, please always use twin-wired cable as your backbone for the whole bus instead of using conduction bars, as indicated in Figure 7.8 .

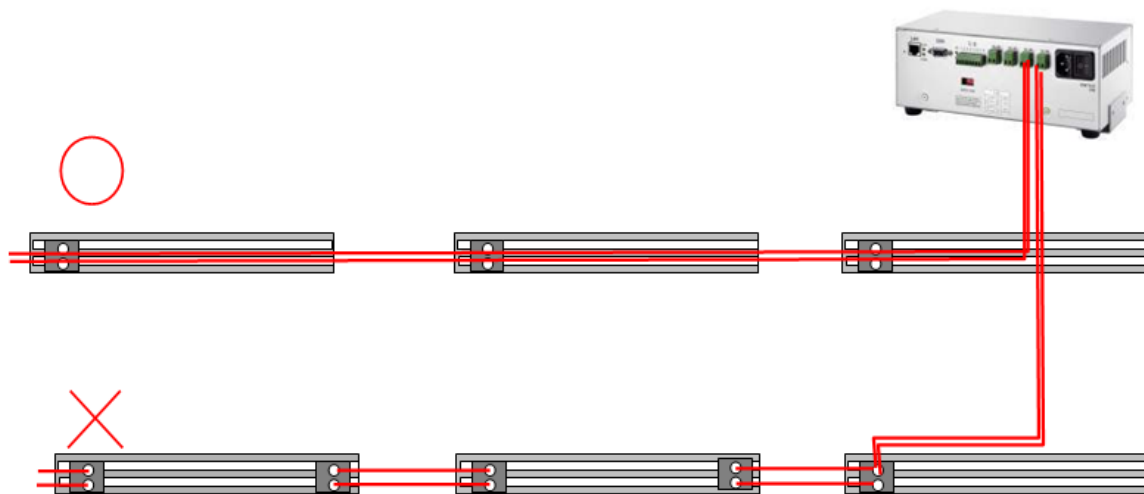


Figure 7.8

7.5 Loading Curve of the Conduction Bar

The line graph below shows the number of the devices that the conduction bar can support with only one terminal block under different lengths. For example, it is guaranteed to support 10 devices with one terminal block in a 6m conduction bar (Figure 7.9 Active region.) However, it would be uncertain to support more than 10 devices, and it would hardly possible to support 21 devices with only one terminal block in the conduction bar (Figure 7.9 Non-active region.)

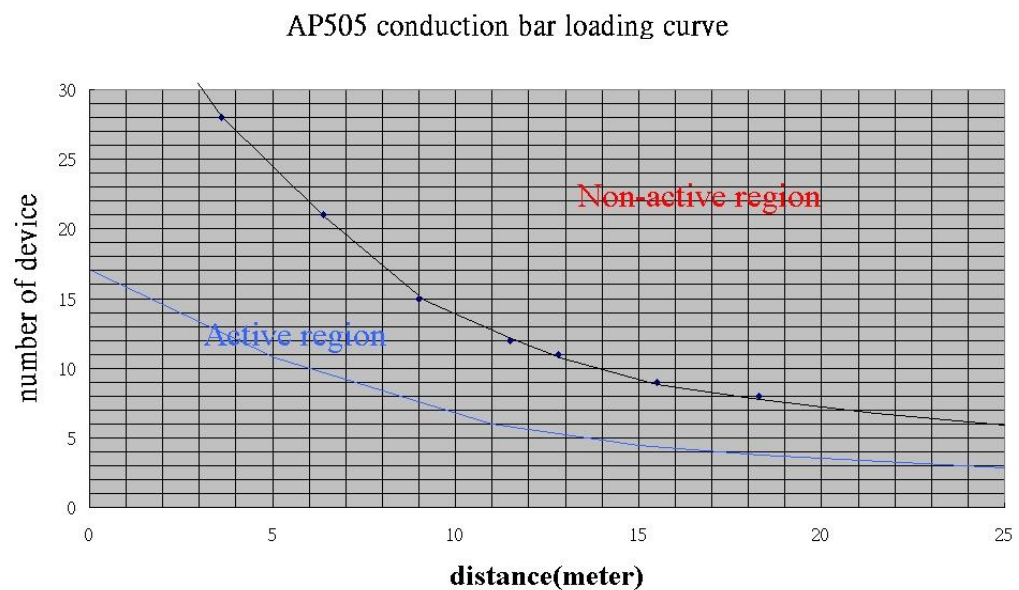


Figure 7.9

7.6 The Crossing-wires Protected by Soft Pipes

Use soft pipes to protect the crossing-wires from shelves to shelves.

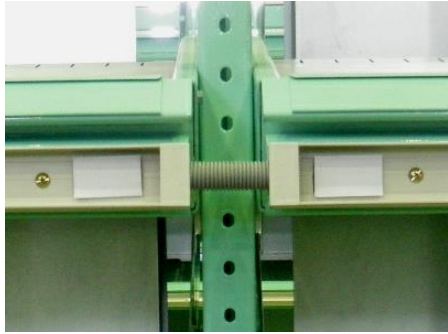


Figure 7.10



Figure 7.11

8 Picking Tags

8.1 Picking Tags Installing

After sticking all the terminal blocks wired by the twin-wired cables, you can put all the devices (picking tags) on. Make sure each of the magnets on the back of the tag is properly stuck on the parallel stainless steel strip on the conduction bar.



Figure 8.1

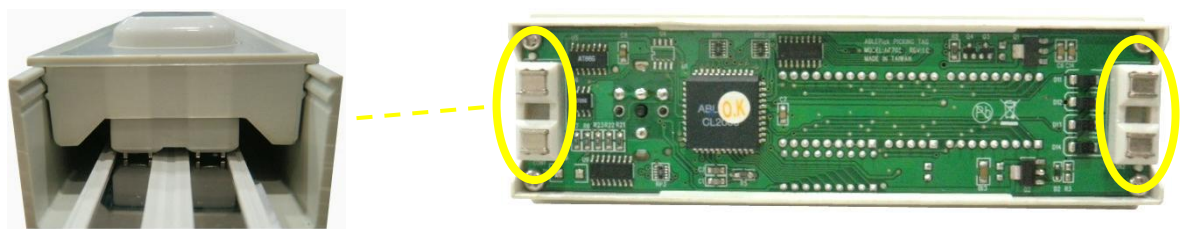


Figure 8.2

Make sure each of magnets is properly stuck on the stainless steel strip on the conduction bar.

8.2 Testing

Once the picking tags are powered on, the lights will be on, and the LED display will start displaying “8.8.8.8.8.8.”, the version of the tags (e.g. U1.0), the default node address “001” and default tag mode “115” orderly. If there is a series of devices keeping resetting, it means the electrical shorts might occur. You will need to check the whole wiring of the buses in the PTL system again to check if they are properly connected with correct polarities. If there is one picking tag not working well, please check if the four magnets on the back are contacted on the stainless steel strips properly.

9 Rail Covers Installing

9.1 Rail Covers

- Material: Plastic (PVC)
- Dimension: L:2000 x W:46(mm)



Please cut the rail covers into size and fill all the gaps between tags to tags.

10 Network Setting

10.1 IP Configuration for the Controllers

As soon as the controller is connected to the internet, the user can proceed to navigate through its configuration by using Monitor. Some information such as the IP, MAC address, etc is going to be displayed, please refer to Figure 10.1. The factory default IP for the controller is 10.0.50.100. The IP address of the PTL controlling PC and the controller on the local area network must be within the same IP address range and subnet mask. Please select its IP by a single-click and press “Config” in the array above. A popping-out window, Dialog, will prompt you to enter your IP address, Gateway and Subnet mask (Mask). After setting, click “Config Now” to complete this process.

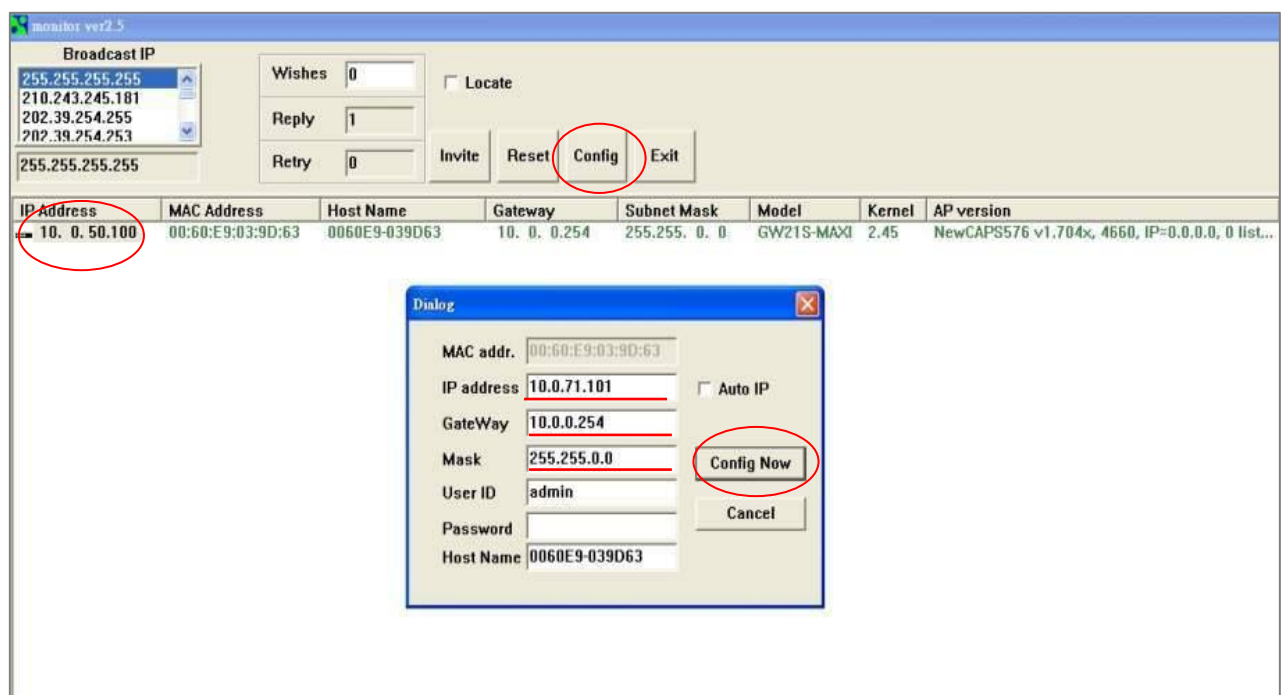


Figure 10.1

11 Tool Introduction

11.1 VB_Tagtest

VB_Tagtest is a software tool allowing you to do the polling range setting, hardware functions inspecting, the node address configuring, and some of the custom setting for the devices under a controller. It is also able to execute an emulation to test the system performance of data-receiving-and-sending.

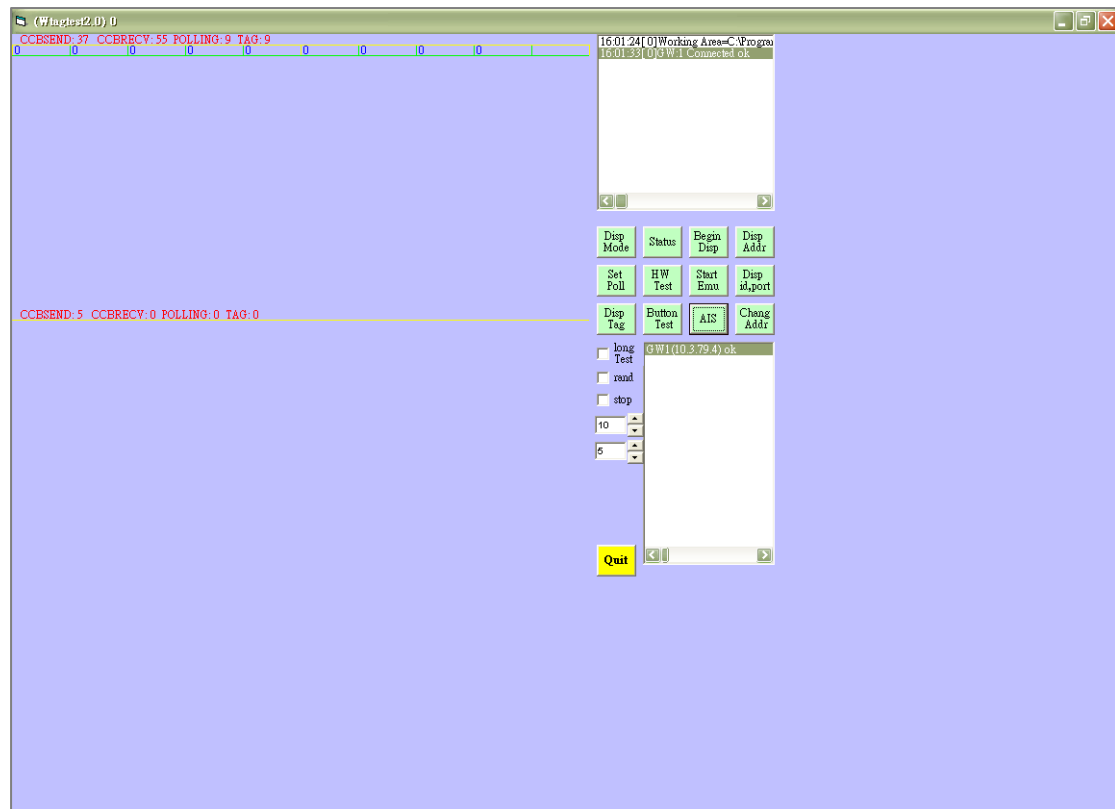


Figure 11.1

REMARK: Please refer to “VB_tagtest.exe User Manual” for more information

12 Node Configuration

Each device should have a unique node address which cannot be duplicated under a controller. The factory default node address for all kinds of picking tag is 001. It can be configured by the hardware, software (AIS) or an address installation tool-AT540. Please make an arrangement to assign node addresses orderly in advance for the picking tags on the shelves with consecutive numbers instead of using disordering numbers.

12.1 Hardware

This will be only suitable for those picking tags which have two function keys, one confirmation button and one LED display, such as AT705, AT506-3W-123, etc.

12.1.1 Buttons on the Devices

Please turn on the power of AT500 to start configuring a new node address for the picking tag by pressing and holding the two function keys for about a second. Press the confirmation button immediately after it. When released, the LED display will show “001”, the factory default. You will need to select a number from 0 to 9 for the blinking digit indicating as a cursor from the far left by pressing up and down function keys. Press confirmation button to move the blinking cursor to the next digit to select a number, and to the next digit to select a number for it, too. After selecting numbers, please have the confirmation button double-pressed to complete node configuration.

Note: If you are using a two-digit picking tag. Please refer to “ABLEPick Picking Tag User Manual” for more information.

12.2 Software

12.2.1 IPINDEX

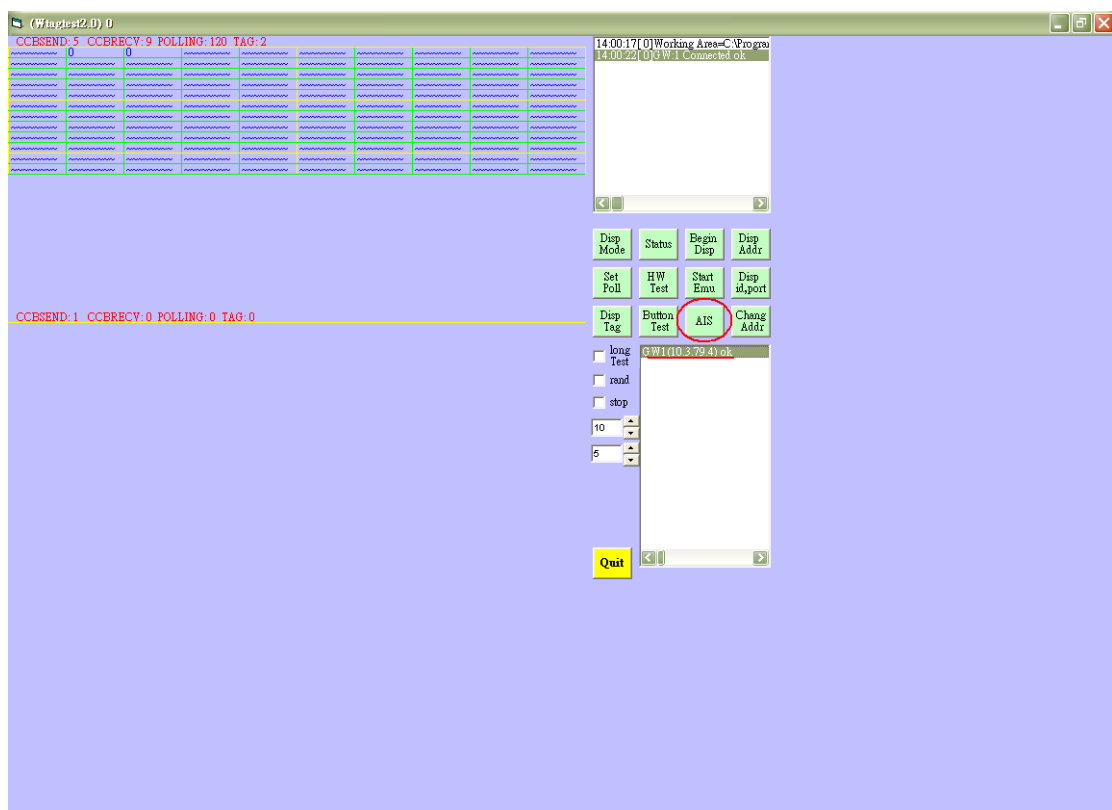
Before configuring the addresses for the picking tags, please save the new IP address of the controller in the “IPINDEX” as a database for the VB_tagtest.



Figure 12.1

12.2.2 VB_tagtest- AIS

When the internet is connected, AIS will allow you to configure the node address for each device under each controller. You can select a controller (gateway) and start assigning node address for the devices under it by pressing the confirmation buttons on the devices. Use AIS to set or obtain a best polling range automatically to let the controller communicate with the devices in the range. Start assigning node address from node 001 orderly by pressing the confirmation button of the designated tag, to 002, 003, 004, and so on. You can also send a specific number as the node address to the designated tag by pressing its confirmation button. For more information, please refer to “VB_tagtest.exe User manual.”



12.3 Address Installation Tool- AT540



AT540 is a portable device able to assign a node address to those picking tags which do not have any function keys and LED display on them. While the AT540 is powered up, the LED light will indicate solid-red. Please put the device with the four magnets on the stainless steel stripes, and boot the AT540 up by pressing and holding AT540's confirmation button for a second. After hearing a "beep" with "AIT 1.0" displayed, please release the button immediately to let the AT540 search for the device (showing "-----" with blinking-red LED.) When the search is done, the device's node address will be shown with a solid-yellow and solid-red LED. Then you can select a node address from 001 to 200 by pressing up and down keys on the AT540 to complete the configuration.

13 Verification

13.1 Node Address

It is suggested to make a plan for node-address-assigning in advance with consecutive numbers instead of using disordering numbers.

1. After setting a best polling range for the controller, please check the status column in the VB_tagtest to see if there is any cell showing “~~~~~” which means the picking tag corresponding with it cannot be found by the controller. If there is, please confirm that the node addresses are assigned properly according to your plan. Please be noticed that a node address should not be double-assigned under a controller.
2. Have all picking tags display their node address simultaneously by using “Disp Addr” in VB_tagtest. Please confirm if the node address shown on each of them corresponds with your previous arrangement.

13.2 Seven-segment Display Testing

A seven-segment display is made up of four vertical and three horizontal segments. It indicates as the Arabic numeral, 8, when it is shown in a full segment. Send “88888” to have the LED display in a full segment and full digit by entering it in Value/Disp Tag/ VB_Tagtest. Confirm if each of the segments is working well.

13.3 Button Testing

To test the function of the buttons, click “Button test” to have all the picking tags display “88888”. After pressing the confirmation buttons, “CCCCC” should be displayed as responses, and “c.c.c.c.c.” should be shown as response after pressing the down key.

13.4 LED Color Testing

Select a color for the designated devices in LedColor/ Disp Tag from red, green, orange, blue, purple, to indigo or select “all” to display all of the colors alternatively.

13.5 Polling Range

Set or obtain the best polling range automatically in “Set Poll” or “HW Test” in VB_tagtest for the controller. Once the polling range is set, it will have the controller communicate with the devices only within the range. For example, if the polling range is set as 120 under 120 devices in a bus. The controller will communicate with each of the devices in order from node 001 to node 120. But if the polling range is set as 100, the controller would only process data exchanges from node 001 to node 100, and node 101 to 120 would be missed. And if the polling range is set as 150, the controller will communicate the devices orderly from node 001 to node 150. It will make vain data exchanges from node 121 to node 150. So, the best polling range will be the maximum node address.

13.6 Electricity Testing

13.6.1 “88888” Displaying

Have all the devices display in a full-segment and full-digit by entering “88888” in the blank of “value” in “Disp Tag”. This will make the device have maximum power consumption. If there is a device or a series of devices not working, it means the power is not enough for it or in the area. It usually occurs in the end of a bus.

13.6.2 AT750-Digital Voltmeter

Use AT750-Digital Voltmeter, a portable voltage tester, to test voltage in a specific area by putting it on the conduction bar. Please refer to Table 13-1 to see the indication of the LED lights.

Table 13-1

LED	Voltage	State
Green	9V ↑	High enough to work
Orange	8V~9V	Uncertain to work well
Red	8V ↓	Too low to work

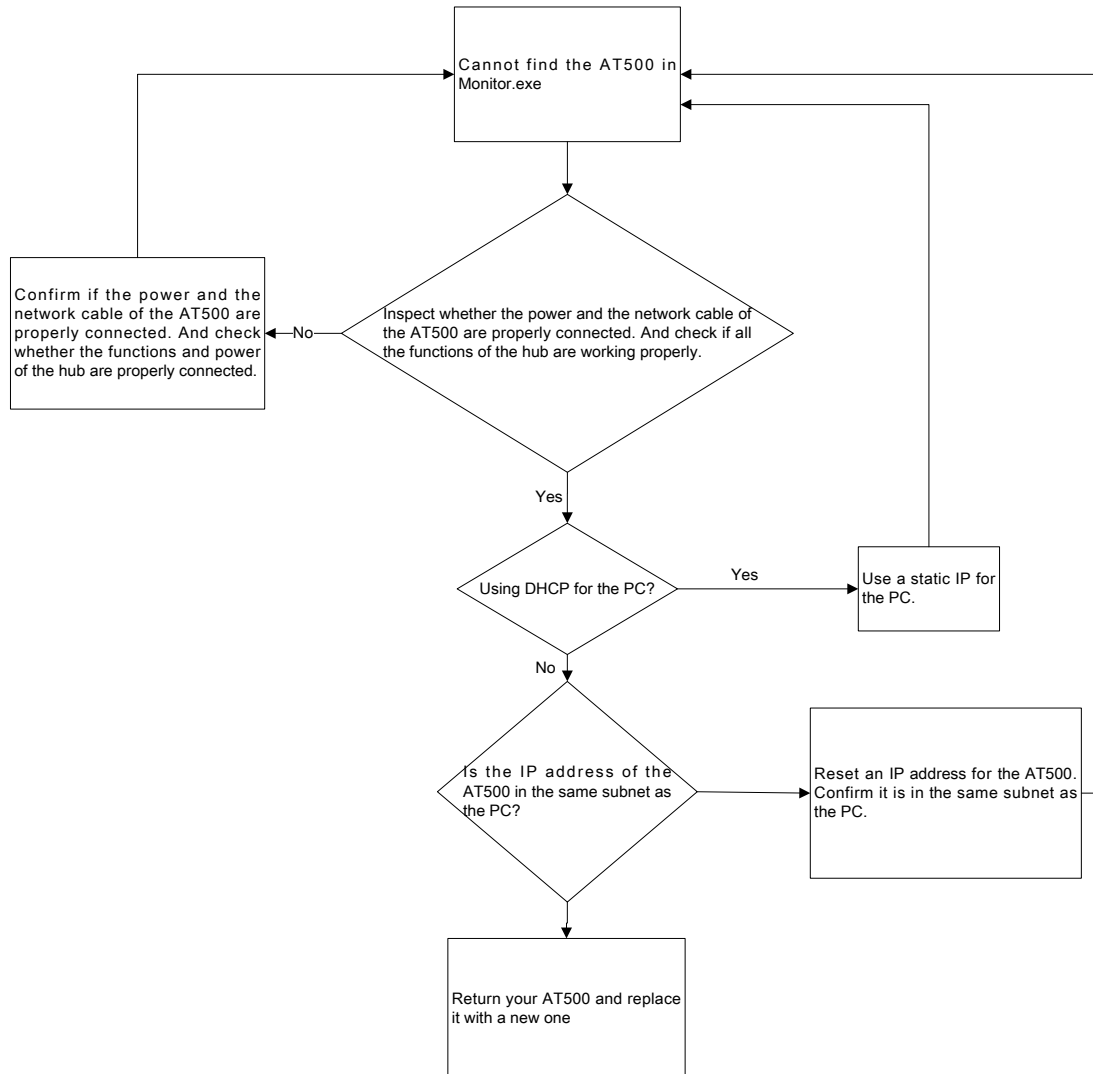
13.7 Emulation Testing

Have the system execute data-receiving-and-sending by clicking “Start Emu” in VB_tagtest to test the system performance.

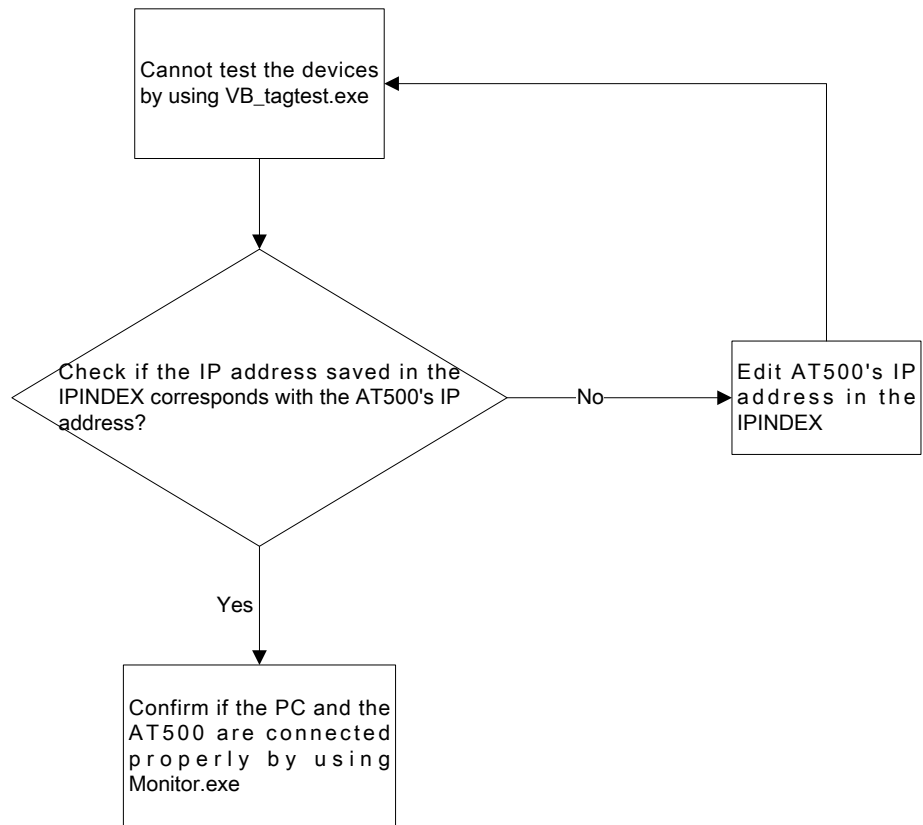
14 Trouble Shooting

14.1 Connection between PC and the AT500

14.1.1 Monitor.exe



14.2 VB_Tagtest



14.3 Problems in the Buses

14.3.1 Devices Keeping Resetting

You will have to check the wiring of the whole buses again when you encounter the situations listed below. Please confirm if the terminal blocks are properly wired with correct polarities and firmly stuck on the conduction bar.

1. The devices are keeping resetting: When an electrical short occur, the power of the AT500 would be cut off and boot up again and again.
2. Unstable communication

14.3.2 Devices Out of Work

If there is a device or a series of devices not working, it might be because of the voltage is not enough in the section. It usually occurs in the end of a bus.

14.4 Problems in the Picking Tags and Controllers

14.4.1 Picking Tags

If there are more than one picking tags lighted up when you send a command to a picking tag, you will need to check whether the node addresses are properly assigned. It is very important to assign a unique node address to a picking tag. Using a duplicated node address to more than one device under a controller would cause picking errors.

Note: Please check if the communication performance, the seven-segment display, the LED lights, and the buttons of the devices are functional. If one of those mentioned above is malfunctioning, please contact us to return and replace it with a new one.

14.4.2 Controllers

1. When an electrical short occur, the power of the controller would be cut off and boot up again and again. The devices under the controller would keep resetting. Please check the whole wiring in the PTL system again to prevent the controller from being damaged by the shorts.
2. If there is one or a series of picking tags showing “FF”, it means there is no connection between the controller and the devices. Please reset a best polling range for the controller.
3. Please always make sure the fan in the controller is working well to prevent the controller from crashing due to overheating.

Note: Please confirm if the controller’s communication performance, the power, output channels, LAN port and all the internal elements are functional. If not, please contact us to return and replace it with a new one.